## HDMI EXTENSION BOARD – XEM8310 CONNECTIONS, POWER AND MOUNTING

1. Function: ties VREFP (MC2.79) to GNDADC (MC2.77), which selects the FPGA internal ADC reference.

2. Source / changes: Opal Kelly XEM8310 Hardware Design Guide, Selecting the ADC Reference Voltage, revision not stated: https://docs.opalkelly.com/xem8310/hardware-design-guide/ BRK8310-A01 SYSMON Header R6 = 0 ohm is a direct wire here. ADC readings are not verified on hardware.

Function: XEM8310 connectors provide the board power, FPGA I/O, JTAG and GT connections; J4 is the 12 V input and MH1–MH4 mount the module.

Source / changes: Opal Kelly BRK8310–A01 p.1 (2022–05–19): https://pins.opalkelly.com/downloads/442/download?category_id=3 ; Codex assigned the custom HDMI and x4 nets. VIO1 target is 1.8 V: program and verify it. V24: J4 changed from CUI PJ–102AH (5 A) to Same Sky PJ–063AH (8 A, 24 V, same 2.0 mm pin and 5.5 mm plug, no switch pin; unused PJ–102AH pin 3 dropped): https://www.sameskydevices.com/product/resource/pj–063ah.pdf ; footprint = KiCad library BarrelJack_CUI_PJ–063AH_Horizontal, pin 1 at the old pin 1. Mating plug and 12 V adapter must also be rated above 5 A; footprint not re–measured against the drawing.

## FRONTEND 5 V POWER | CUSTOM MICRO–HDMI PORTS – NOT STANDARD HDMI

Function: U700 converts +VDCIN to FE_5V (5.02 V nominal, 6 A module rating).
PG and R703 enable all eight ports once the central rail is in regulation.

Source / changes: TI TPSM63606 SLVSGB4B Rev. B, Fig. 9–1 and Table 8–1,
https://www.ti.com/lit/ds/symlink/tpsm63606.pdf . Codex adapted prior V14 U100:
R700 40.2k, R701 10k, R702 13k (all 1%), C708 22pF set 5 V / 1 MHz; PG gates switches.
C700–C703 10uF / 50V; C704–C707 use 47uF / 16V X5R instead of prior X6S.
CBOOT/RBOOT shorted, VLDOIN tied to output. Check effective capacitance,
loop stability, input damping, startup, thermal performance and PCB layout.

Function: U701–U708 individually switch FE_5V to J610–J617 pin 19; red LEDs indicate each switch FAULT. TP701–TP708 access the outputs. All ports start together; no staggering.
Source / changes: TI TPS2553 SLVS841G Rev. G, pins / sections 8–9 and Table 9–2: https://www.ti.com/lit/ds/symlink/tps2553.pdf . Codex adapted prior V14: EN now follows U700 PG;
R711–R718 = 37.4k 1% (630–771 mA limit); C711–C718 are IN bypass; R721–R728 / D701–D708 added as FAULT lights. Each OUT goes to a separate pin–19 net.
Cable current and return rating, hot–plug / ESD, backfeed, 8–port inrush and thermal behavior are not verified. Existing 3.3 V–only frontends must NOT be connected to these 5 V ports.

## PCIe x4 – BRK8310 reference circuit

Function: Connects PCIe x4 data and 100 MHz reference clock, with reset, SMBus and WAKE#. Host slot 12 V is isolated from the board 12 V input.

Source / changes: Opal Kelly BRK8310–A01 schematic, p. 2, 2022–05–19: https://pins.opalkelly.com/downloads/442/download?category_id=3
Adapted from x8 to x4: fit R28 = 0 ohm; omit R29 and unused lanes 4–7 and C21–C26/C29–C30. R2/R7 and R9–R11 retained.
Unused options R1/R8, R16/R17 and R27 are omitted.
Host +12 V goes only to TP20. Host 3.3 V and AUX power the original sideband circuitry; they do not feed the main 12 V input.
PCA9306 Rev O, Fig. 8–1, section 10: https://www.ti.com/lit/ds/symlink/pca9306.pdf ; SN74AXC1T45 Rev E: https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
AP22653 DS41186 Rev 5–2, sections 7–9: https://www.diodes.com/datasheet/download/AP22652_AP22653_AP22652A_AP22653A.pdf
Q1 Si1922EDH dual FET, product datasheet: https://www.vishay.com/en/product/67192/
Working target: VIO1 = 1.8 V (Howard, 3 Nov 2026; needed for PCA9306 >= 1.2 V and Bank 64 LVDS): set in XEM before testing and verify all Bank 64 I/O. Factory default 1.0 V is below PCA9306 range: https://docs.opalkelly.com/xem8310/device–settings/
ERC power flags at U2/U6 VREF2 represent their R13/R23 200k resistor–fed bias, not independent regulators. Confirm bias and startup.
PCIe/HDMI combined Vivado design, clock, VIO1 choice, host sequencing and signal integrity remain to be verified.

## USB–C JTAG PROGRAMMER, SELECTION AND FRONTEND PORTS

## USB TO JTAG PROGRAMMER

Function: J500 and U501 convert USB–C to JTAG; U505 stores the programmer settings.
U602 translates the JTAG lines to 1.8 V. SW600 selects the XEM or one of eight frontends.
U603 checks the selected target voltage; U605 enables the translator after a delay.
C601/C603 bypass the XEM rails; U610 selects the XEM path. Turn OFF before selecting.

Source / changes: USB–C input: Codex adaptation of BRK8305 A03,
25Oct2023, JTAG sheet8/PDFp4: https://pins.opalkelly.com/downloads/583/download?category_id=3
J500=USB4110–GF–A; D500=USBLC6–2SC6. C506
changed 10nF to 100nF. USB shell uses C502/FB503;
shared DGND. Connector fit and USB/ESD layout unverified.
ST USBLC6–2 Rev7 Fig15: https://www.st.com/resource/en/datasheet/usblc6–2.pdf

FT232H and EEPROM: Codex adaptation of BRK8305 A03,
25Oct2023, JTAG sheet8/PDFp4: https://pins.opalkelly.com/downloads/583/download?category_id=3
FT232HQ pins13/14/15/16=TCK/TDI/TDO/TMS; ADBUS4–7
and ACBUS NC as BRK8305. Former R508 removed.
U505=93LC56BT–I/SN (SOIC) replaces CFG001/MSOP;
Y500=SJK replaces Abracon; FB500–502=Murata BLM18AG601SN1D.
Y500 specifies 18pF load; 18pF C503/C504 give ~9pF + stray.
Confirm crystal load, frequency, drive and startup after layout.
Y500: https://xonstorage.z8.web.core.windows.net/pdf/sjk_7u12000e18ucg_xonjuly20_20_link.pdf
Bead current and voltage drop need checks.
Microchip DS21794F Table3–1: https://ww1.microchip.com/downloads/en/DeviceDoc/21794F.pdf
FT232H DS v2.2 sec3.6: https://ftdichip.com/wp–content/uploads/2024/09/DS_FT232H.pdf
AMD program_ftdi EEPROM write/readback and Vivado
enumeration must be tested. VCK190 match not established.
https://docs.amd.com/r/2024.1–English/ug908–vivado–programming–debugging/Programming–FTDI–Devices–for–Vivado–Hardware–Manager–Support

Translator: Codex adaptation; TI TXU0304 Rev A sec.9: https://www.ti.com/lit/ds/symlink/txu0304.pdf
VCCA = FTDI3V3; VCCB = XEM +1.8 V (MC1.2); R601–R603 = 22R provisional.
R604 pull–up removed; IC inputs have weak pulldowns. Check rail sequencing, unpowered–target
injection, timing, cable loading and FTDI/Vivado operation.

Rail bypass: Codex adaptation: C601/C603 retained from the removed regulators.
XEM8310 MC2.70 = +3.3 V and MC1.2 = +1.8 V; rail budget and startup must be verified.
Opal Kelly XEM8310 pin list: https://pins.opalkelly.com/pin_list/XEM8310

Rotary selector: Codex direct–to–enable wiring; nine 10k pulls replace decoder logic.
ALPS SRBQ 1P9T non–shorting; confirm stock, contact sequence and footprint.
https://tech.alpsalpine.com/e/products/detail/SRBQ490100/
Bounce and switch faults are not proven.

OFF/ENABLE: Codex dual–pole arrangement; R600 biases MASTER low.
C&K JS datasheet p.9; verify contact positions and footprint:
https://www.littelfuse.com/assetdocs/littelfuse–c–k–slide–js–series–datasheet?assetguid=aba42b08–0d2c–423b–813d–a2faa5a3bb14

Voltage window: Codex circuit; TI TPS3700 Rev G sec.7.3: https://www.ti.com/lit/ds/symlink/tps3700.pdf
XEM 3.3 V supplies U603. Selected VREF uses XEM R629 or HDMI pin 1.
R605–R608=301k/100k and 402k/100k; R613/C608 discharge/filter.
Nominal window 1.604–2.008 V; verify leakage, tolerance and startup.

Delay: Codex circuit; TI TPS3808 Rev N secs.4, 7: https://www.ti.com/lit/ds/symlink/tps3808.pdf
U605 changed G33 to G30: 2.79 V nominal threshold increases 3.3 V rail margin.
CT open: nominal 20 ms; R615=12k now pulls to XEM 3.3 V, R616=100k pulls down.
Startup and brownout remain unverified; keep OFF during power–up.
JLC C189211 listing conflicts with TI G30 threshold; verify MPN/marking.

XEM target switch: Codex circuit using TI TS3A27518E Rev F Fig.5–2, Table 8–1:
https://www.ti.com/lit/ds/symlink/ts3a27518e.pdf
IN1/IN2 high select NO; active–low EN from rotary; 10k pullup. Channels (Howard 1 Nov 2026, for straight PCB routing): TCK=4, TDI=6, TMS=3, TDO=5, VREF=2; channel 1 unused.
NO2–NO6: ±6 kV IEC contact rating; not a 5 V DC clamp.
Selection, leakage, power–off and SI need checks.
XEM JTAG pins: https://docs.opalkelly.com/xem8310/expansion–connectors/

CUSTOM FRONTEND INTERFACE – NOT HDMI VIDEO

### FRONTEND PORT 1 | JTAG SWITCH + MICRO–HDMI

Function: HDMI1 selects shared JTAG and VREF.
100R series parts alone do not provide 5 V protection.

Source / changes: Codex circuit using TI TS3A27518E Rev F Fig.5–2, Table 8–1:
https://www.ti.com/lit/ds/symlink/ts3a27518e.pdf
IN1/IN2 high select NO; active–low EN from rotary; 10k pullup. Channels (Howard 1 Nov 2026, for straight PCB routing): TCK=4, TDI=6, TMS=3, TDO=5, VREF=2; channel 1 unused.
NO2–NO6: ±6 kV IEC contact rating; not a 5 V DC clamp.
Selection, leakage, power–off and SI need checks.

Function: HDMI 1 carries recording, command, JTAG, sense and 5 V.
U801 helps shunt data–line ESD; shell is bonded to DGND.

Source / changes: Codex custom interface; Molex 467651001 SD–46765–001 sheets 1–2:
https://www.molex.com/content/dam/molex/molex–dot–com/products/automated/en–us/salesdrawingpdf/467/46765/467651001_sd.pdf?inline=
XEM connector pin list: https://pins.opalkelly.com/pin_list/XEM8310
J610 3/5 to MGT_RXP0_224/MGT_RXN0_224; 12/14 to B64_L1P/B64_L1N.
U801 TI TPD4E02B04 SLVSD85B Rev B secs. 5, 6.6, 8.2 and 10:
https://www.ti.com/lit/ds/symlink/tpd4e02b04.pdf
U IO1/2/3/4 = HDMI 3/5/12/14; flow–through pads 10/9/7/6 tied to IO1/2/3/4
(SLVSD85B p3 pin table, straight–through routing); both GND pins to DGND.
Direct shell bond assumes a common–ground bench system.
Connector, cable, SI, ESD layout/strike and enclosure need verification.

### FRONTEND PORT 2 | JTAG SWITCH + MICRO–HDMI

### FRONTEND PORT 3 | JTAG SWITCH + MICRO–HDMI

### FRONTEND PORT 4 | JTAG SWITCH + MICRO–HDMI

### FRONTEND PORT 5 | JTAG SWITCH + MICRO–HDMI

### FRONTEND PORT 6 | JTAG SWITCH + MICRO–HDMI

### FRONTEND PORT 7 | JTAG SWITCH + MICRO–HDMI

### FRONTEND PORT 8 | JTAG SWITCH + MICRO–HDMI

V24 one–page schematic: J4 = PJ–063AH

Sheet: /
File: HDMI_Extension_Board_Study_V24.kicad_sch

**Title: HDMI Extension Board – XEM8310 and USB–C JTAG**

Size: User
Date: 2026–11–03
**Rev: V24**
KiCad E.D.A. 10.0.6
Id: 1/1